Evan Cummings

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Education

Ph.D. Biomedical Engineering, Imaging and Ultrasonics Track **2025**
University of Michigan, Ann Arbor, Michigan
Thesis title: *Rosette Trajectory Cardiac Magnetic Resonance Fingerprinting*
Thesis Advisor: Nicole Seiberlich

M.S. Biomedical Engineering, Imaging and Ultrasonics Track **2021**
University of Michigan, Ann Arbor, Michigan

B.S. Biomedical Engineering, Devices and Instrumentation Track **2019**
Case Western Reserve University, Cleveland, Ohio

Research Experience

Seiberlich Lab/Michigan Institute for Imaging Technology and Translation **2017-2026**

Rosette MR Fingerprinting for T_1, T_2, T_2*, and Fat Fraction Mapping (2020-2025)

- Developed a trajectory, sequence, and reconstruction pipeline to simultaneously measure T_1, T_2, T_2*, and fat fraction maps from a single-breathhold sequence to improve non-invasive quantitative assessment of disease
- Techniques reduced required breathhold duration, improved spatial resolution, and reduced quantitative errors compared to previous approaches.
- Optimized acquisition and reconstruction techniques for cardiac, liver, and carotid applications across 0.55T, 1.5T, and 3T systems
- Investigated iterative and deep learning-based techniques to improve reconstruction quality
- Presented findings at ISMRM Annual Meeting (2021, 2022, 2024, 2025), Low-Field Workshop (2022), RSNA Annual Meeting (2023), and SCMR Annual Meeting (2025)
- Manuscript accepted for publication in Magnetic Resonance in Medicine

Multipoint Reconstruction for Radial GRAPPA (2017-2022)

- Worked with the through-time radial GRAPPA reconstruction algorithm to improve the speed and memory requirements for reconstructing highly undersampled data, giving clinicians more freedom to switch slice orientations during a real-time scan or interventional procedures
- Approach reduced calibration time from 30s to 5s with minimal impact on image quality
- Findings presented at the Annual Meetings of BMES (2017) and ISMRM (2018), and published in Magnetic Resonance in Medicine (2022)

Siemens Healthineers Collaboration/Internship **2023**

MR Fingerprinting Development Kit (MRFDK) (2023)

- Worked with the Siemens development team onsite in Erlangen to implement and test 3D prostate and free-running cardiac MRF sequences in the MRFDK framework
- Provided feedback to the Siemens development team for improvement of the MRFDK
- Prepared educational guides and presentations for the MIITT group on the use and potential applications of the MRFDK
- Made a recommendation to Seiberlich lab whether to integrate it into the lab's development workflow

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Technical Skills

- Physics: Proficient with MR Physics and Bloch equation simulation.
- Mathematics: Proficient working with inverse problems and optimization algorithms
- Pulse Sequence Development: Proficient with C++ and Siemens IDEA environment
- Image Reconstruction: Proficient working with parallel imaging, compressed sensing, and machine learning-based reconstruction algorithms in MATLAB, Python, and Tensorflow

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Additional Skills

- Collaborations: Worked with colleagues internally to help adapt my techniques for application in the liver and other organs, worked with external partners on collaborative development of software tools, and worked with clinician scientists to deploy and evaluate techniques in patient studies.
- Communication: Experienced in written communication in the form of prepared abstracts and manuscripts, and have presented research at 6 international conferences.

Publications

Cummings E, Cruz G, Kaplan S, Hamilton J, Seiberlich N. Deep image/deep IDEAL prior reconstruction enables rosette trajectory MR fingerprinting at 0.55T for single-breathhold T_1, T_2, T_2*, and fat fraction mapping. [manuscript in preparation]

Griesler T, Hamilton J, **Cummings E**, Richardson J, Whiteman A, Gram M, Cruz G, Davenport M S, Hussain H, Seiberlich N. Liver T1, T2, T2*, and PDFF mapping at 0.55T using rosette MRF with optimized sequence design and deep image reconstruction. [manuscript in preparation]

Cummings E, Cruz G, Richardson J, Kaplan S, Hamilton J, Seiberlich N. Rosette cardiac MR fingerprinting for simultaneous T_1, T_2, T_2*, and fat fraction mapping using a multi-echo deep image prior reconstruction. *Magn Reson Med* (2026) [accepted for publication]

Ahad J, **Cummings E**, Franson D, Hamilton J, Seiberlich N. Optimization of through-time radial GRAPPA with coil compression and weight sharing. *Magn Reson Med* 88:1244-1254 (2022)

Cummings E, MacDonald JA, Seiberlich N (2021). Parallel Imaging. In Prieto C, Doneya M, Akacaya C, *MRI Reconstruction: Theory, Methods, and Applications.* Elsevier.

Conference Proceedings

Cummings E, Cruz G, Hamilton J, Seiberlich N. Single-breathhold Cardiac T_1, T_2, T_2*, and fat fraction mapping at 0.55T using dual rosette trajectory MRF and a deep image prior reconstruction. *International Society for Magnetic Resonance in Medicine Annual Meeting,* Honolulu, 2025.

Cummings E, Cruz G, Hamilton J, Seiberlich N. Cardiac T_1, T_2, T_2*, and PDFF Mapping at 0.55T with Rosette MR Fingerprinting and a Deep Image Prior Reconstruction. *Society of Cardiac Magnetic Resonance Annual Meeting,* Washington DC, 2025.

Cummings E, Cruz G, Kaplan S, Richardson J, Hamilton J, Seiberlich N. Cluster-based Low-rank Regularization for Cardiac Rosette MR Fingerprinting. *International Society for Magnetic Resonance in Medicine Annual Meeting,* Singapore, 2024.

Cummings E, Cruz G, Liu Y, Jiang Y, Hamilton J, Seiberlich N. Simultaneous Single-breathhold Cardiac T_1, T_2, T_2*, and PDFF Mapping with Rosette MR Fingerprinting. *Radiological Society of North America Annual Meeting*, Chicago, 2023.

Cummings E, Liu Y, Jiang Y, Ropella-Panagis K, Hamilton J, Seiberlich N. Simultaneous Mapping of T_1, T_2, and T_2* at 0.55T with Rosette MR Fingerprinting. *International Society for Magnetic Resonance in Medicine Annual Meeting*, London, 2022.

Cummings E, Liu Y, Jiang Y, Ropella-Panagis K, Hamilton J, Seiberlich N. Simultaneous Mapping of T_1, T_2, T_2*, and Fat Fraction at 0.55T with Rosette MRF. *ISMRM Workshop on Low Field MRI*, Virtual, 2022.

Cummings E, Liu Y, Ropella-Panagis K, Hamilton J, Seiberlich N. Rosette MRF for Simultaneous T_1, T_2, and R_2* Mapping. *International Society for Magnetic Resonance in Medicine Annual Meeting,* Virtual, 2021.

Cummings E, Franson D, Hamilton J, Seiberlich N. Sharing Radial GRAPPA Weight Sets Across k-Space to Decrease Memory Requirements for Real-Time Imaging. *International Society for Magnetic Resonance in Medicine Annual Meeting*, Paris, 2018.

Cummings E, Franson D, Hamilton J, Seiberlich N. Improving Radial GRAPPA Efficiency by Reconstructing Multiple Points from a Single Weight Set. *Biomedical Engineering Society Annual Meeting*, Phoenix, 2017.

Teaching Experience

Teaching Assistant, Modeling of Biomedical Systems **Spring 2019**

- Assisted in teaching Modeling of Biomedical Systems, a Junior-level course that covers basic modeling techniques for a variety of biomedical applications, including drug delivery, cell population dynamics, neuron activation, and musculoskeletal motion.
- Held weekly office hours to assist students and helped run weekly MATLAB-based simulation laboratories.

Teaching Assistant, Biomedical Signals and Systems **Fall 2018**

- Assisted in teaching Biomedical Signals and Systems, a Junior-level course that covers solving differential equations in time and frequency domains, linear system analysis, continuous and discrete Fourier transforms, and basic filter design.
- Planned and ran weekly recitations for around 20 undergraduate students to help them prepare for quizzes and exams in the class.